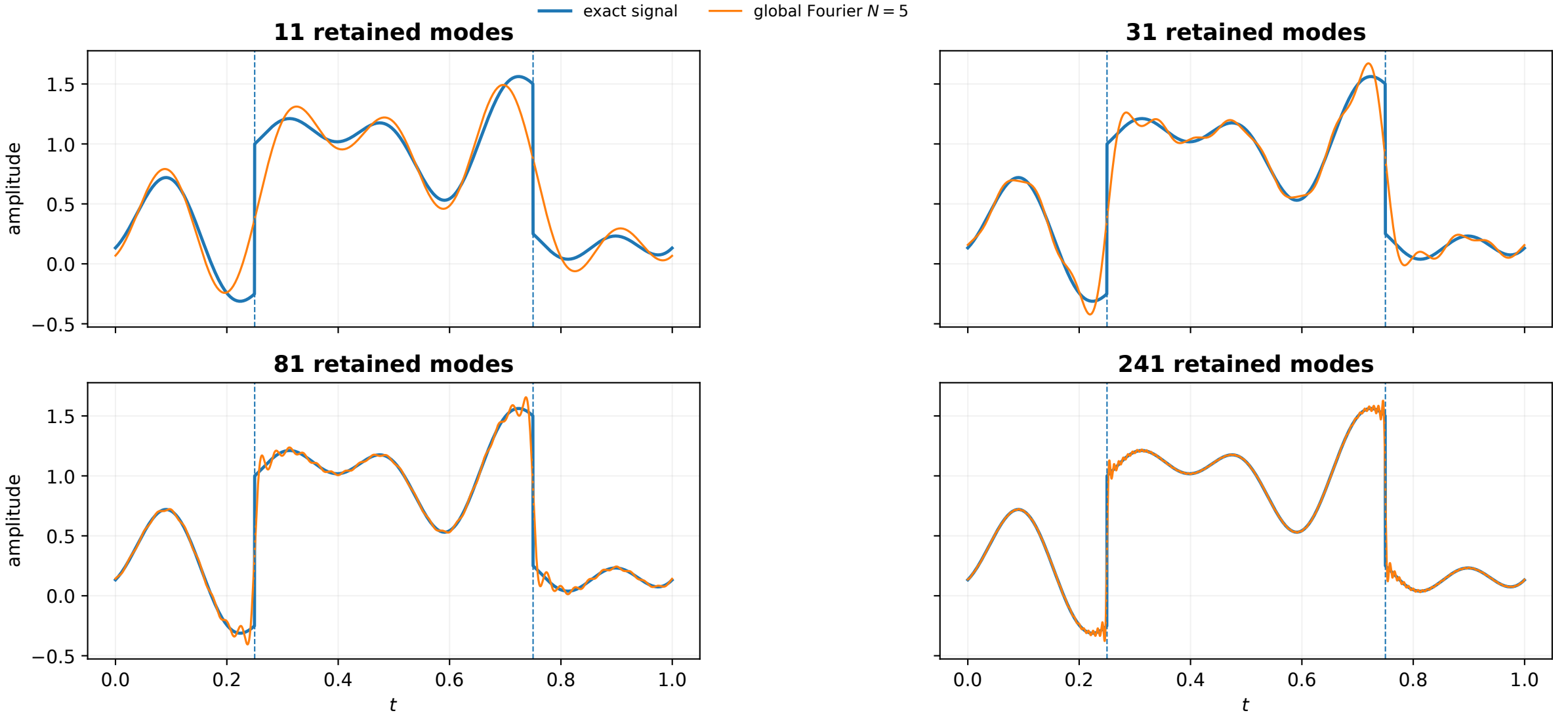Global Fourier resolution narrows the oscillatory region but does not reduce its limiting relative height